

Diagnosing and Forecasting Distribution Shift in Sequential Preference Models

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October 31, 2026

1 Motivation and Research Objective

Pairwise reward models are trained to predict which of two alternatives is preferred. Their deployment is inherently sequential: the distribution of compared alternatives may change, the underlying preferences may evolve, and both processes may interact. A deterioration in prediction quality can consequently have several explanations. New comparisons may expose uncertainty or estimation error in an otherwise unchanged preference rule; the rule itself may have changed; or a covariate shift may precede and contribute to a later preference shift. These mechanisms require different responses, yet can produce similar observed losses.

Recent work studies uncertainty as an early signal of concept drift [4], change-point estimation for pairwise comparisons with covariates [3], and preference drift in dynamic Bradley–Terry models [5]. Reward-model research further shows that good in-distribution error need not prevent failure under distribution shift [6]. This project targets the intersection of these problems. Its objective is not to propose another general change-point algorithm, but to characterize what epistemic uncertainty and outcome-based prediction surprise can diagnose, when they fail, and whether covariate changes can provide advance warning of later preference drift.

The primary research question is:

Under which distribution shifts do uncertainty and prediction surprise provide reliable and distinguishable signals of failure in a sequential preference model?

The final stage asks a prospective question: *when only some covariate shifts are followed by delayed preference drift, can the geometry of the covariate shift forecast which shifts are genuine precursors?*

2 Model and Detection Signals

At time t , alternatives A and B have features $x_{A,t}, x_{B,t} \in \mathbb{R}^d$, with pair difference $z_t = x_{A,t} - x_{B,t} \sim Q_t$. The preference outcome $y_t = 1$ denotes $A \succ B$ and follows the covariate Bradley–Terry model [1]

$$y_t \mid z_t \sim \text{Bernoulli}(p_t^*(z_t)), \quad p_t^*(z_t) = \sigma(w_t^\top z_t), \quad \sigma(a) = (1 + e^{-a})^{-1}. \quad (1)$$

An initial stationary sample is used to fit Bayesian logistic regression. A Laplace approximation gives $w \mid \mathcal{D}_0 \approx \mathcal{N}(\hat{w}, \Sigma)$, and the reference model predicts $\hat{p}_t = \sigma(\hat{w}^\top z_t)$. Its epistemic uncertainty in the latent utility difference is

$$u_t^2 = z_t^\top \Sigma z_t. \quad (2)$$

The label-dependent signal is predictive surprise, measured by log loss

$$\ell_t = -y_t \log \hat{p}_t - (1 - y_t) \log(1 - \hat{p}_t). \quad (3)$$

A one-sided CUSUM [2] monitors persistent excess loss, $S_t = \max\{0, S_{t-1} + \ell_t - \mu_0 - k\}$, and alarms when $S_t > h$. A rolling-accuracy monitor provides a deliberately simple baseline. Thresholds will be calibrated to a common pre-change false-alarm rate so that detection delays are comparable.

The controlled experiments separate a change in Q_t with fixed w_t (covariate shift), a change in w_t with fixed Q_t (preference shift), and simultaneous changes. Covariate shifts will vary in their alignment with well- and poorly identified directions of the training design. This is essential: a covariate shift need not raise uncertainty, and increased uncertainty need not imply harmful error. Indeed,

$$\mathbb{E}_{Q_t}[u_t^2] = \text{tr}(\Sigma \text{Cov}_{Q_t}(z)) + \mathbb{E}_{Q_t}[z]^\top \Sigma \mathbb{E}_{Q_t}[z], \quad (4)$$

which exposes the geometric conditions under which a change in Q_t affects the uncertainty signal.

3 Hypotheses, Final Forecasting Stage, and Evaluation

The first hypothesis is that uncertainty responds primarily to covariate movement into weakly identified directions, whereas its association with future loss depends on whether those directions are also relevant to the model’s estimation error. The second is that a pure preference shift with unchanged Q_t cannot be detected by any label-free statistic of z_t ; after labels arrive, log-loss CUSUM should detect persistent disagreement more efficiently than rolling accuracy. The third is that the joint path (u_t, ℓ_t) can diagnose some shift classes but cannot identify all causes without further structural assumptions. The project will state these identifiable and non-identifiable cases explicitly rather than treating every alarm as evidence of preference drift.

The final experiment introduces delayed, endogenous drift. A covariate change occurs at τ_c and, for a subset of events, changes the preference parameter after a lag L , so that $w_{\tau_c+L} = w_0 + \Delta w$; other covariate shifts are benign, while separate preference shifts occur without a covariate precursor. With $\Delta\mu = \mathbb{E}_{Q_{\text{after}}}[z] - \mathbb{E}_{Q_{\text{before}}}[z]$, a simple exposure model sets $\Delta w = \alpha B \Delta\mu$, where the fixed low-rank map B represents preference susceptibility and is specified independently of the detector. Shifts in the null space of B , including selected covariance-only changes, are benign. Across repeated streams, the uncertainty-weighted change

$$R_u = |\mathbb{E}_{Q_{\text{after}}}[z^\top \Sigma z] - \mathbb{E}_{Q_{\text{before}}}[z^\top \Sigma z]|$$

will be compared with an unweighted moment-shift score for predicting whether $\|\Delta w\|$ exceeds a fixed threshold within horizon H . Sweeping the alignment between B and the uncertain eigendirections of Σ tests when uncertainty contains genuine precursor information. This yields a testable early-warning task while avoiding a tautological rule that defines drift directly from the detector score. If the pre-drift observations have the same distribution for benign and preference-inducing events, forecasting is impossible; this serves as a formal negative control and clarifies the assumptions required for positive lead time.

Performance will be assessed by false-alarm probability, detection delay, and lead time relative to a defined degradation in expected preference loss. Because the data-generating parameters are known, the analysis will distinguish irreducible Bernoulli entropy from excess log loss due to model mismatch. The forecasting stage will additionally report discrimination of precursor versus benign shifts and lead time to the subsequent change in w_t . Parameter sweeps will be restricted to shift magnitude, geometric alignment, preference-change magnitude, and lag. The intended contribution is a reproducible synthetic characterization, supported by small analytical results, of when uncertainty warns, when feedback is indispensable, and when a covariate shift contains information about future preference drift. Although deliberately model-simple, the setting retains the pairwise structure used by modern reward models and directly complements recent work on uncertainty-aware reward modeling and non-stationary preference optimization [5, 6].

Selected References

- [1] R. A. Bradley and M. E. Terry. Rank analysis of incomplete block designs: I. The method of paired comparisons. *Biometrika*, 39(3/4):324–345, 1952.
- [2] E. S. Page. Continuous inspection schemes. *Biometrika*, 41(1/2):100–115, 1954.
- [3] Y. Han and T. C. M. Lee. Change point detection in pairwise comparison data with covariates. *arXiv:2408.13642*, 2024.
- [4] P. Lu, J. Lu, A. Liu, and G. Zhang. Early concept drift detection via prediction uncertainty. *Proceedings of the AAAI Conference on Artificial Intelligence*, 39(18):19124–19132, 2025.
- [5] S. Son, W. Banks, S. Ray Chowdhury, B. Paige, and I. Bogunovic. Right now, wrong then: Non-stationary direct preference optimization under preference drift. *Proceedings of the 42nd International Conference on Machine Learning*, PMLR 267:56173–56203, 2025.
- [6] L. Fluri, L. Lang, A. Abate, P. Forré, D. Krueger, and J. M. V. Skalse. The perils of optimizing learned reward functions: Low training error does not guarantee low regret. *Proceedings of the 42nd International Conference on Machine Learning*, PMLR 267:17306–17377, 2025.